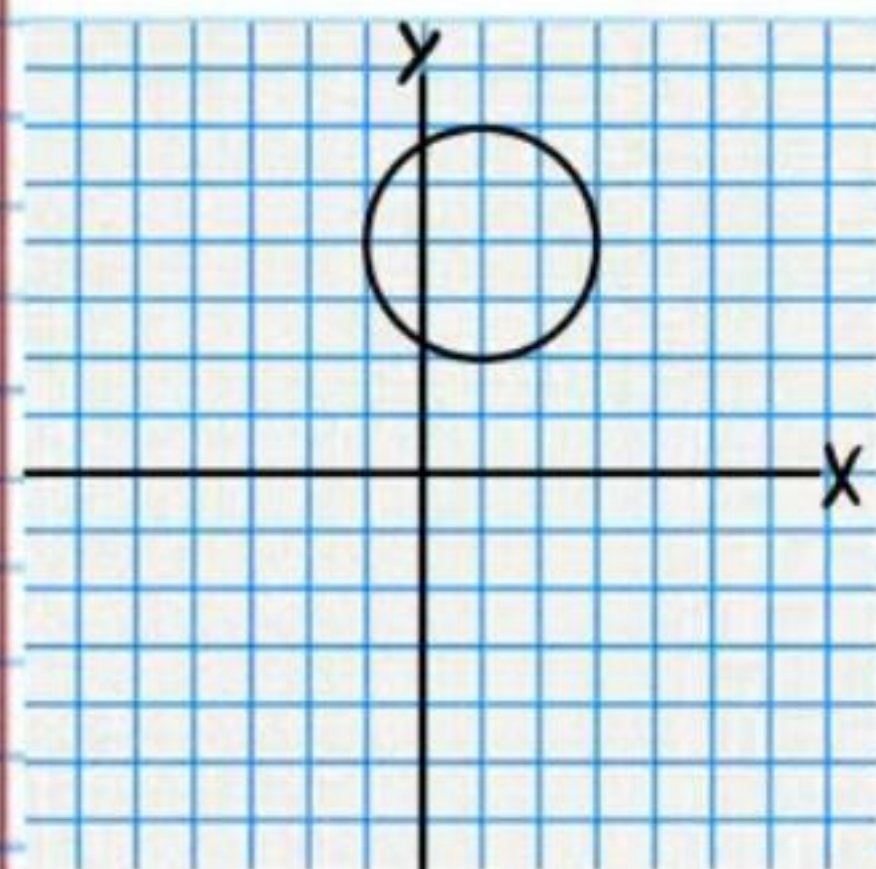


Geometry Test V

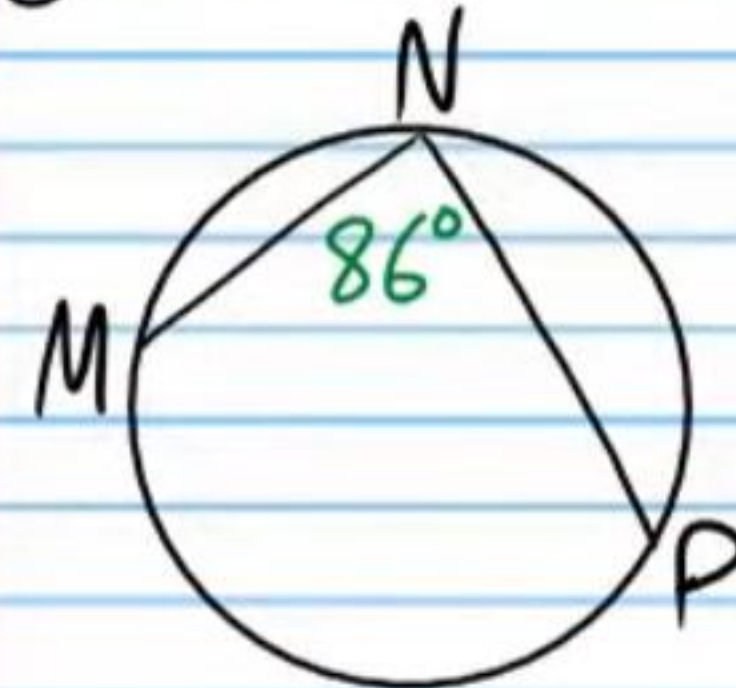
Name
Date
Course

- I) Write your name, the date, and the course title (i.e. Geometry).
- II) Show work when necessary.
- III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
- IV) You will need a calculator for some problems on this test. When pi is required, use the π button on your calculator.
- V) Graphing paper may be useful for this test.
- VI) Each problem is worth four points for a total of 100 points.

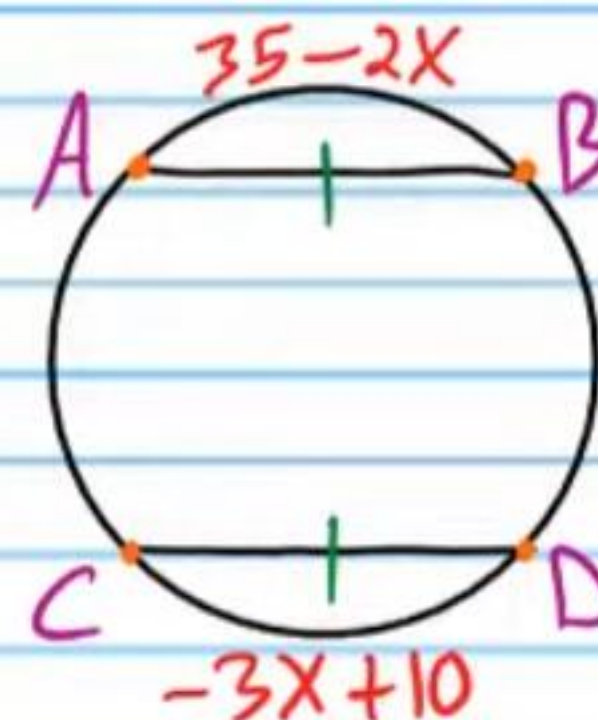
① Write an equation of the following circle.



② Find $m\hat{MP}$

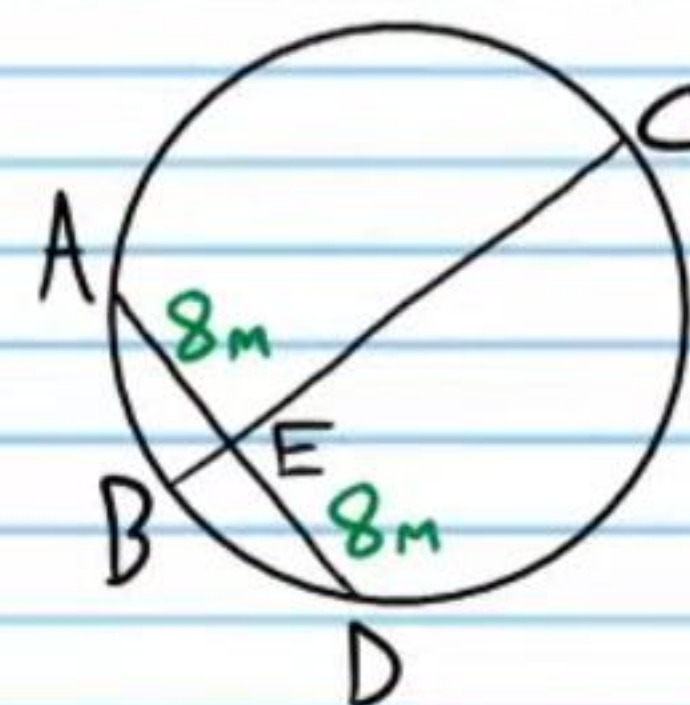


③ Find $m\hat{AB}$

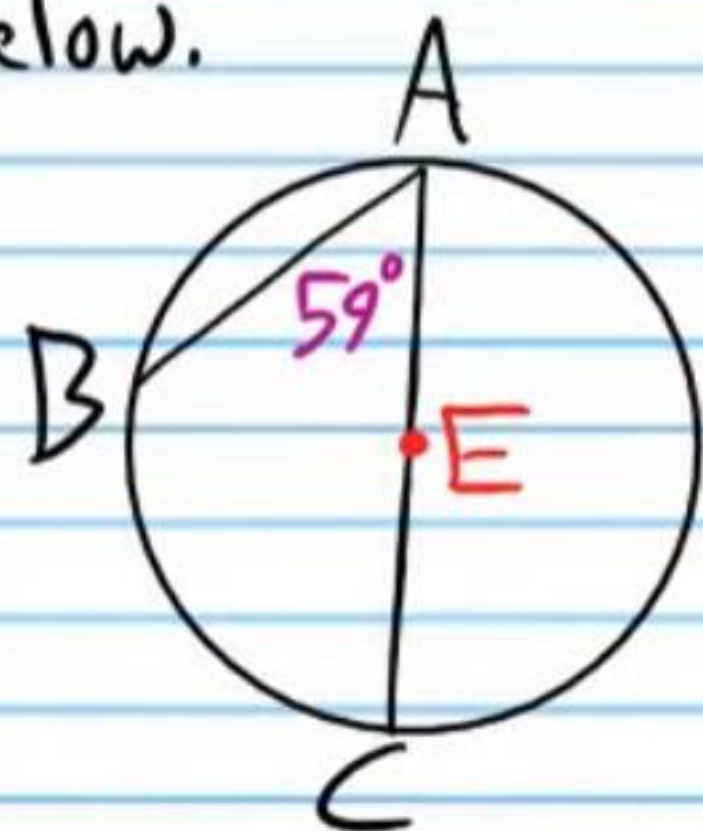


④ Is $\overline{BC}$ a diameter? Explain.

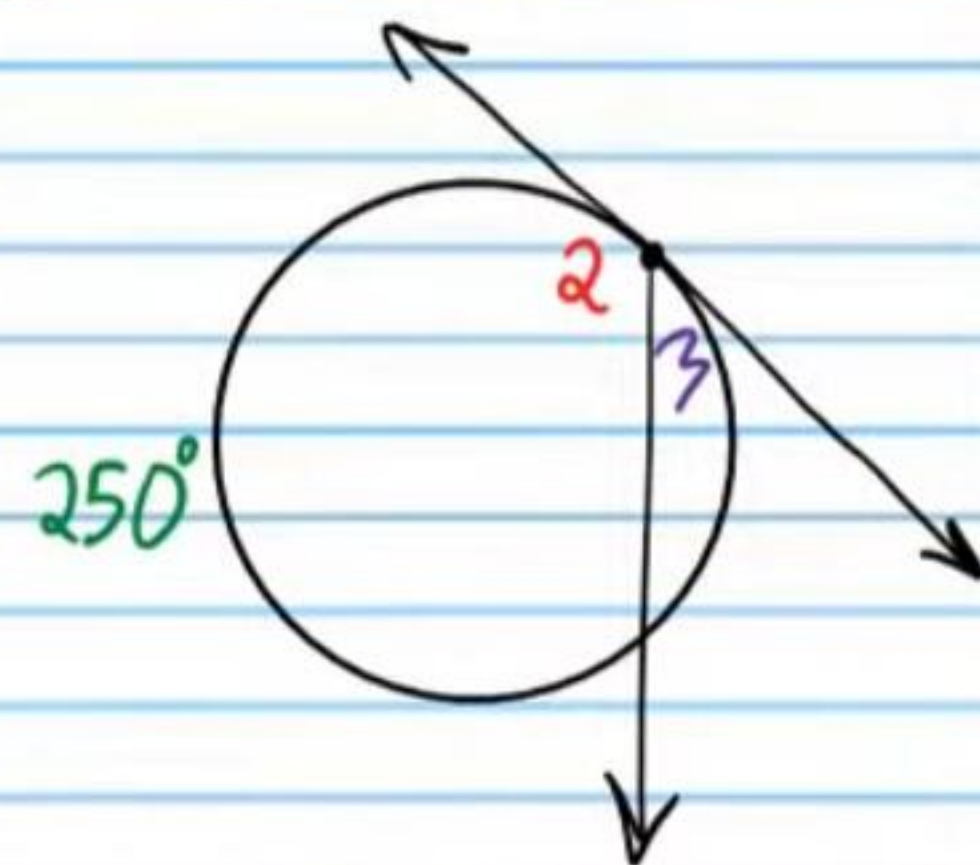
$$\star m\angle DEC = 91^\circ$$



⑤ Find $m\hat{AB}$ if E is the center of the circle below.

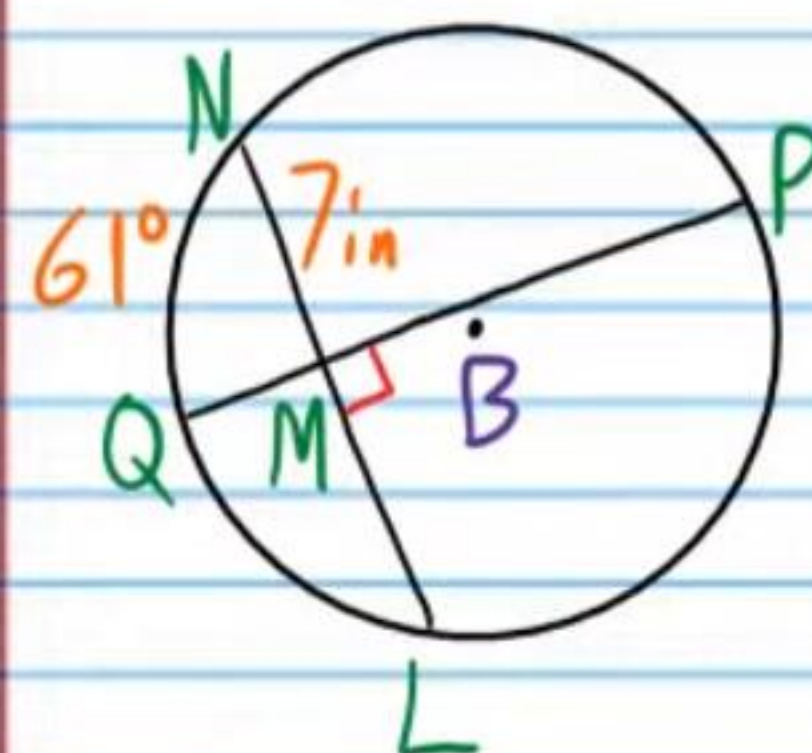


⑥ Find $m\angle 3$.

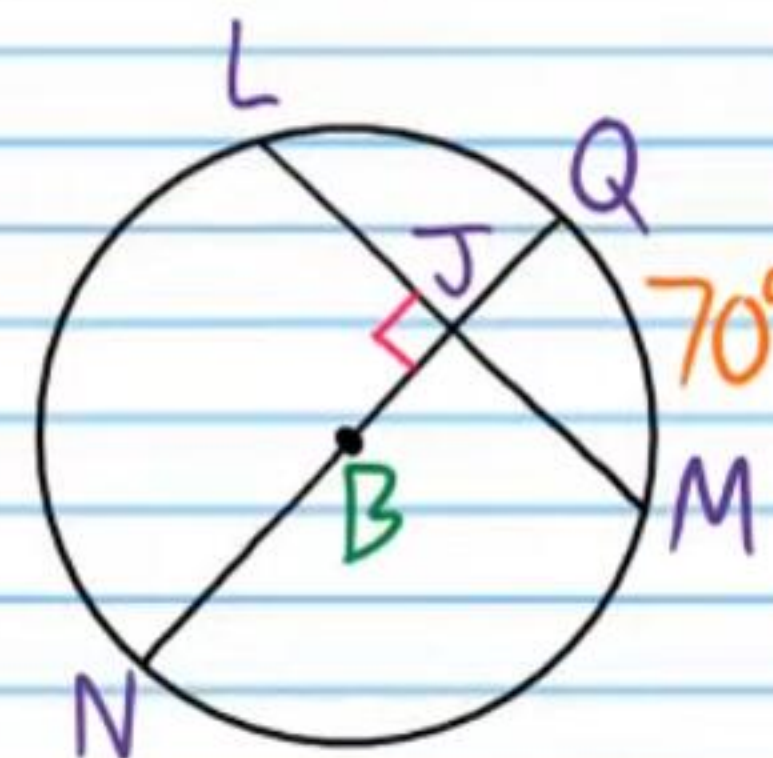


If B is the center of the circles below, find the following values.

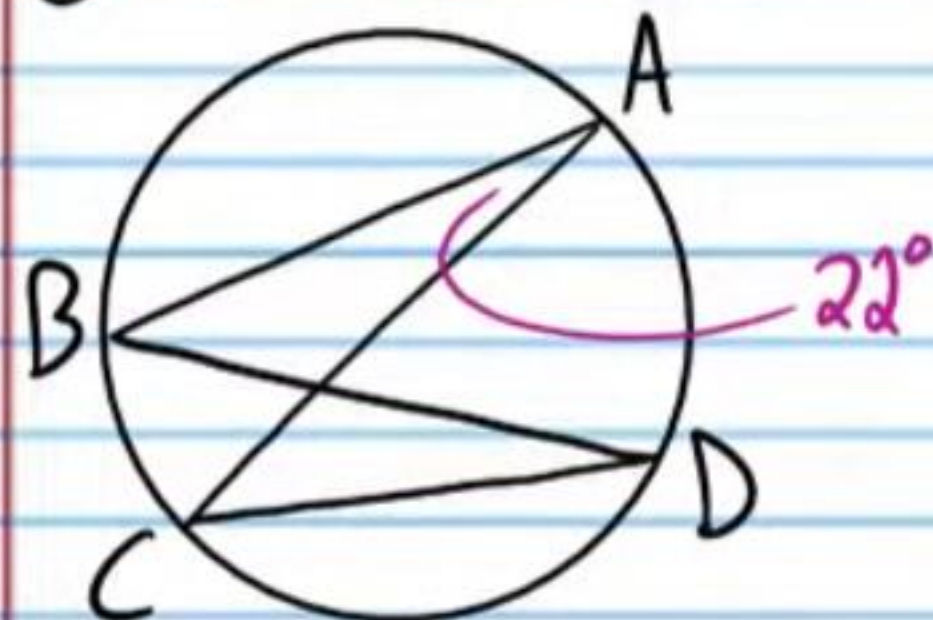
⑧ NL & $m\hat{QL}$



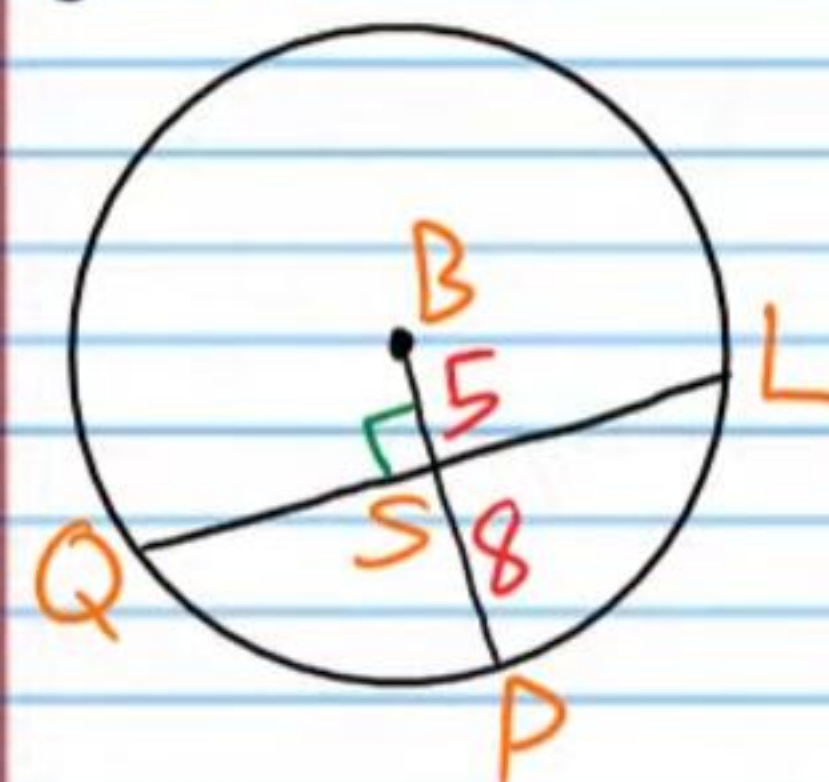
⑨ $m\hat{LQM}$



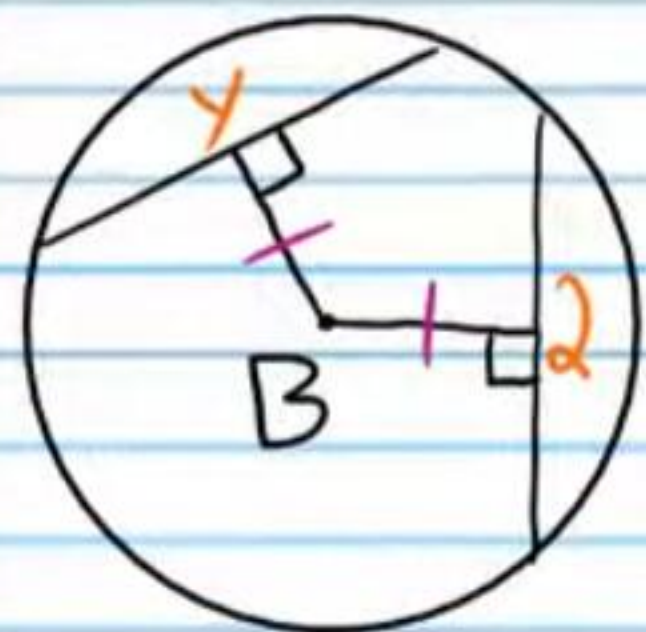
⑦ Find $m\angle D$



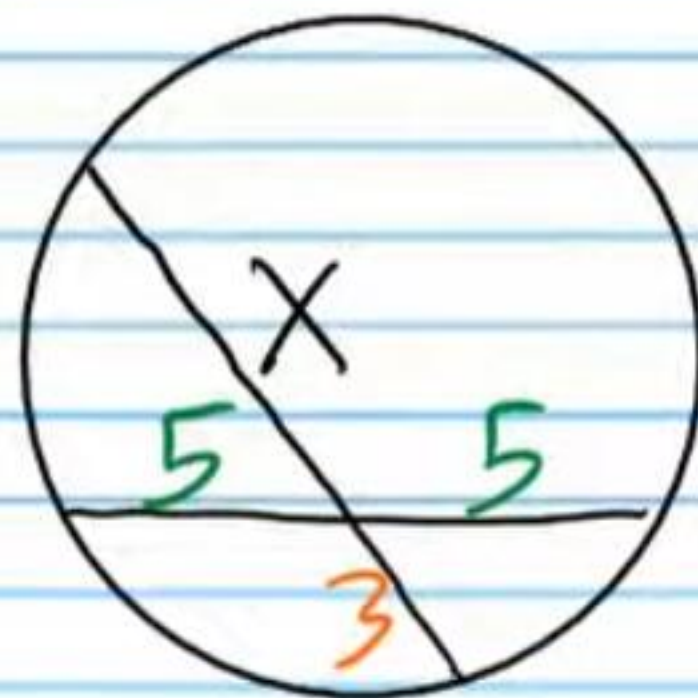
⑩ QS



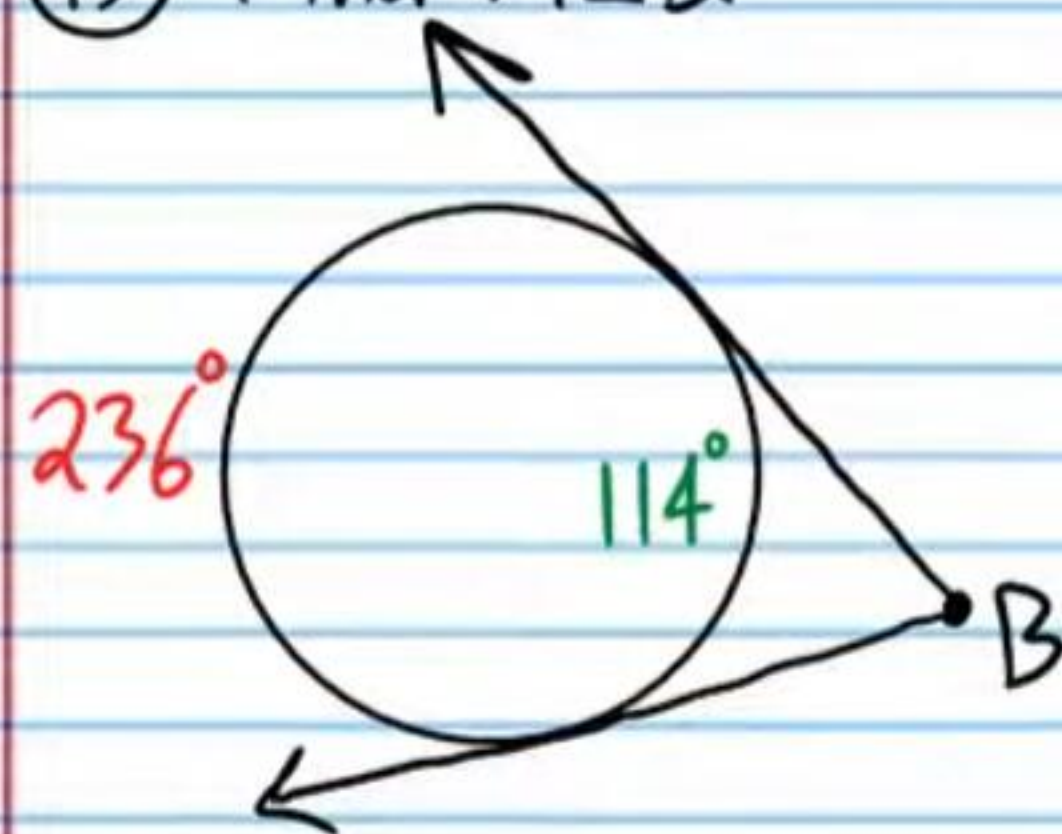
⑪ If B is the center of the circle below, find y.



⑫ Find X.



⑬ Find $m\angle B$.

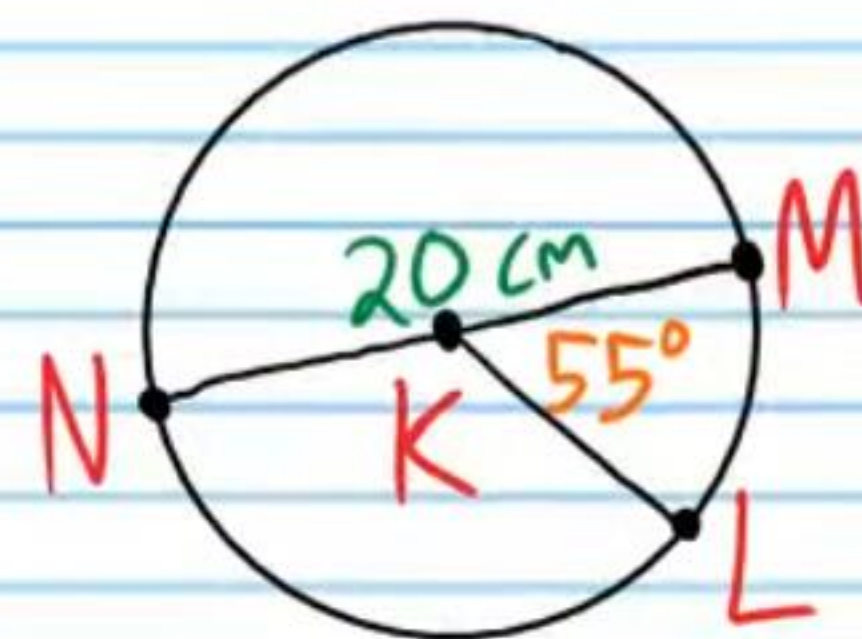
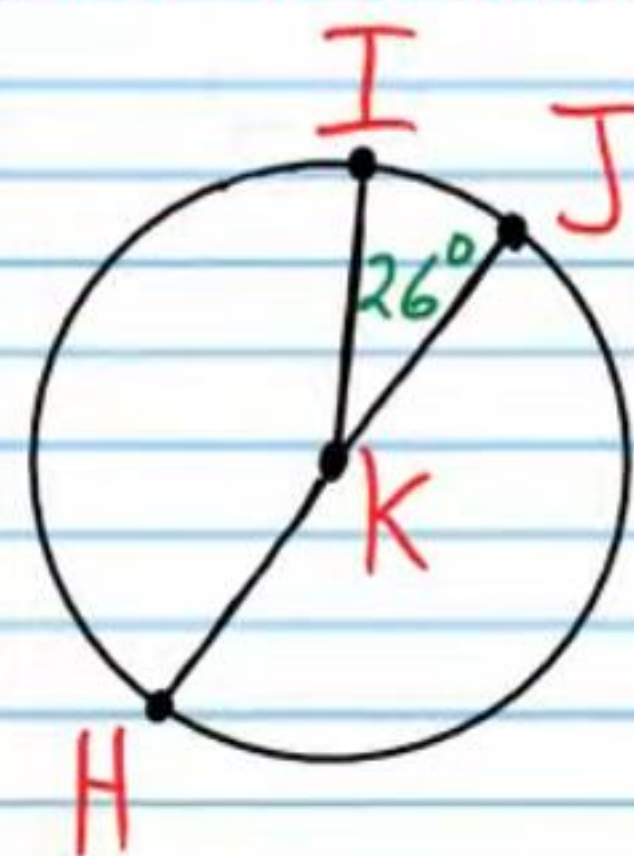


If K is the centers of the following circles, find the values below. If a problem requires the pi value, use the π button on your calculator. Round your answers to the nearest tenth if necessary.

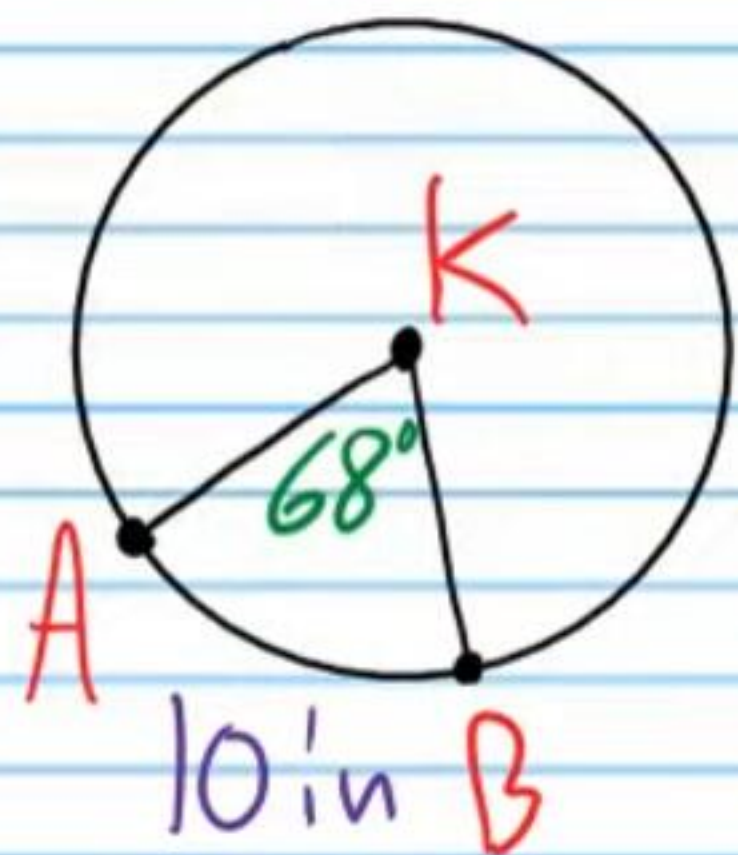
⑭ i) $m\widehat{HKI}$ ii) $m\angle HKJ$ ⑮ $\angle \widehat{LN}$

* $\overline{HJ}$ is a diameter.

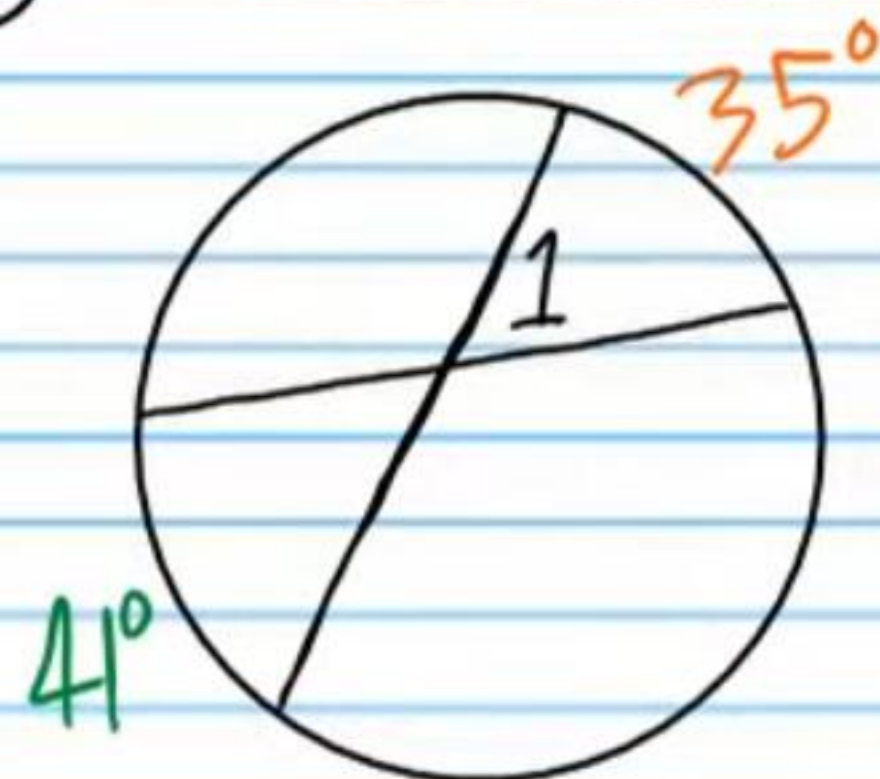
* $\overline{NM}$ is a diameter.



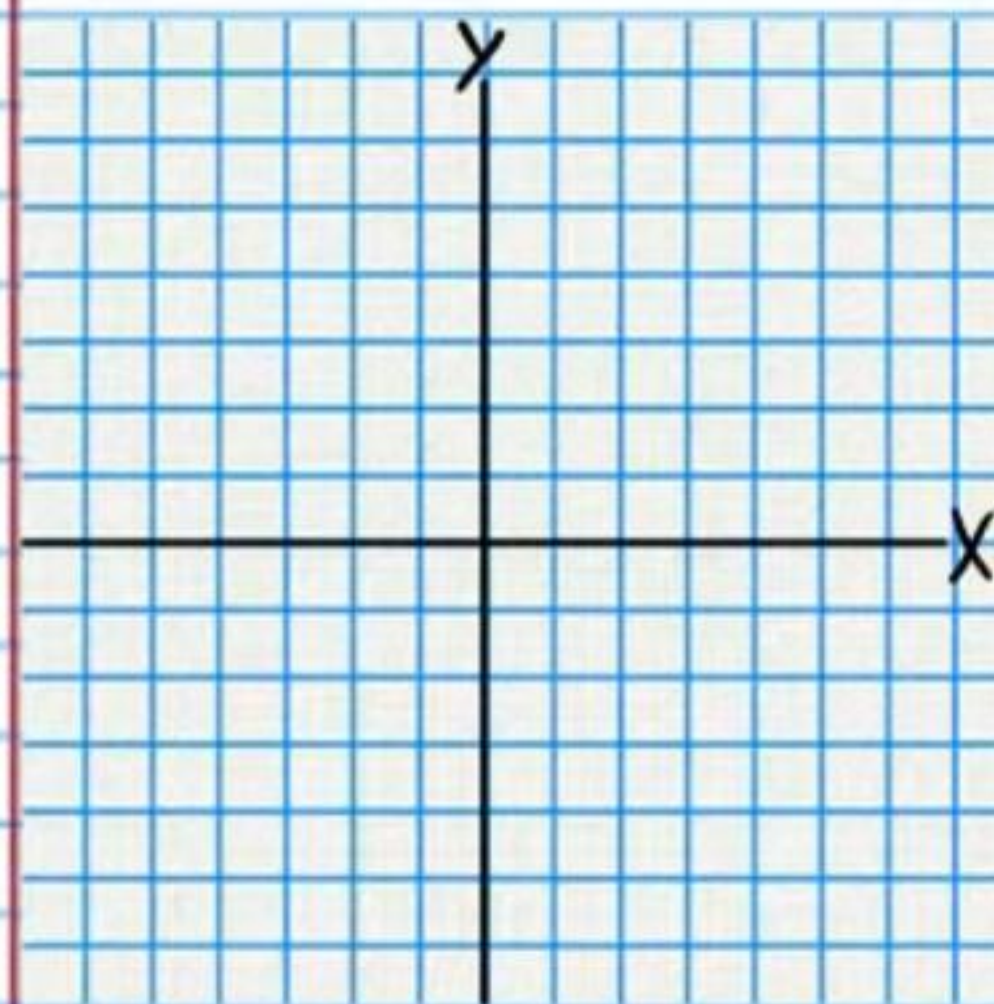
⑩ Diameter



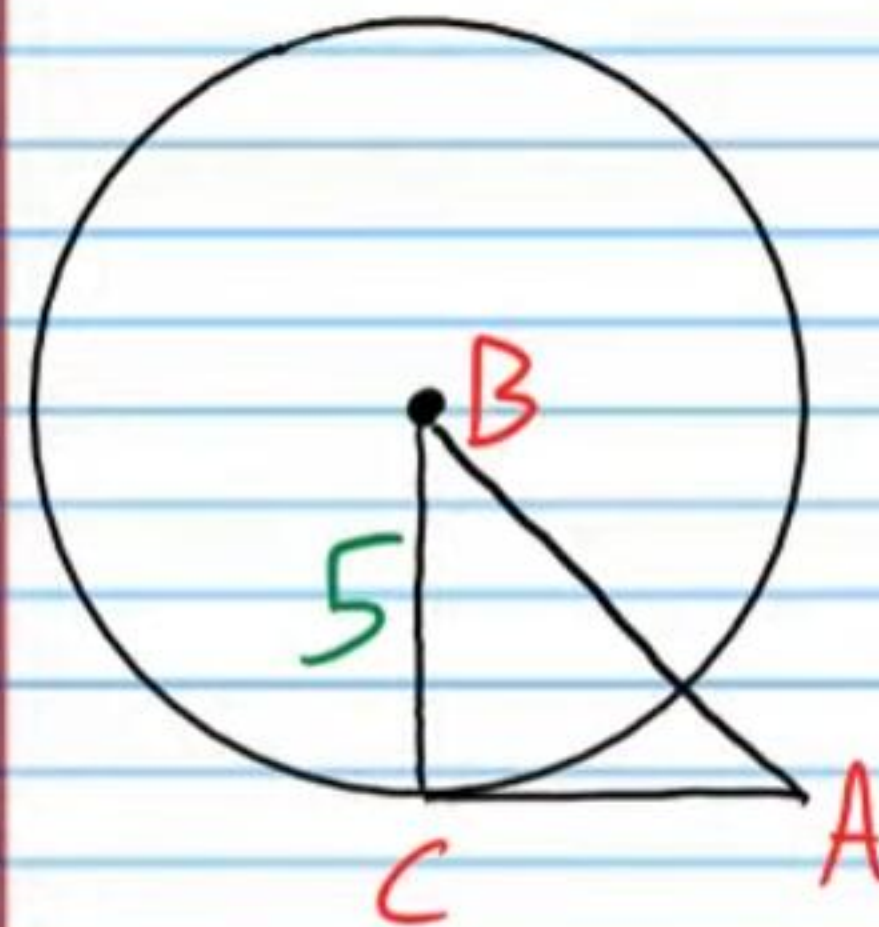
⑪ Find $m\angle 1$.



⑫ Graph the following equation: $(x+3)^2 + (y-1)^2 = 16$.

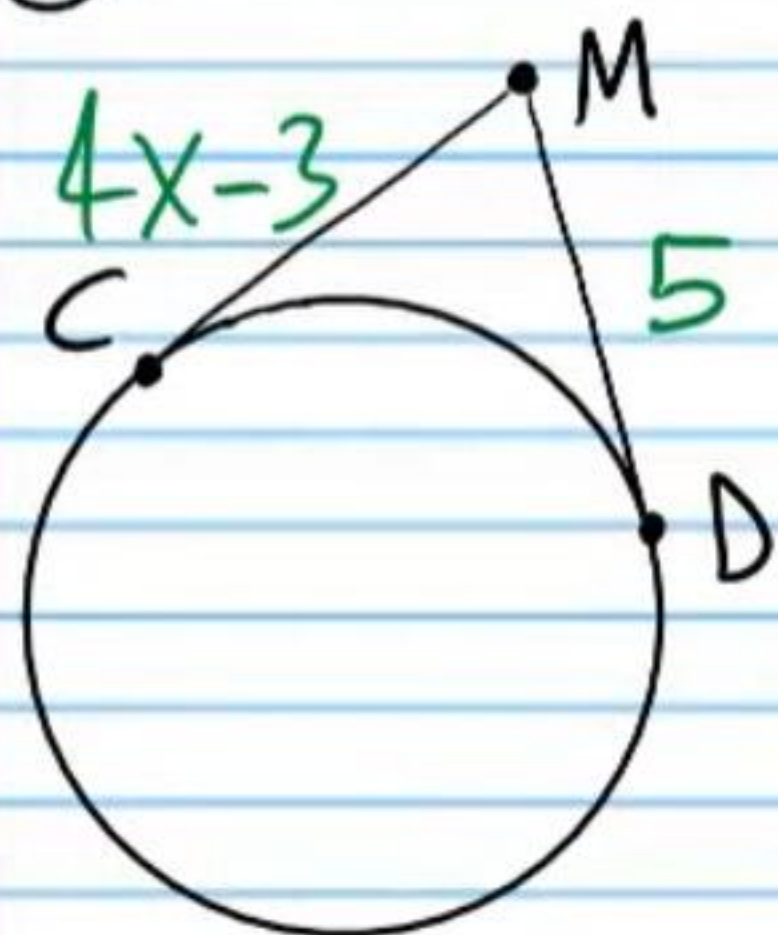


⑬ If the line containing $\overline{AC}$ is tangent to $\odot B$ at C, and $AB = 5\sqrt{2}$, find AC.

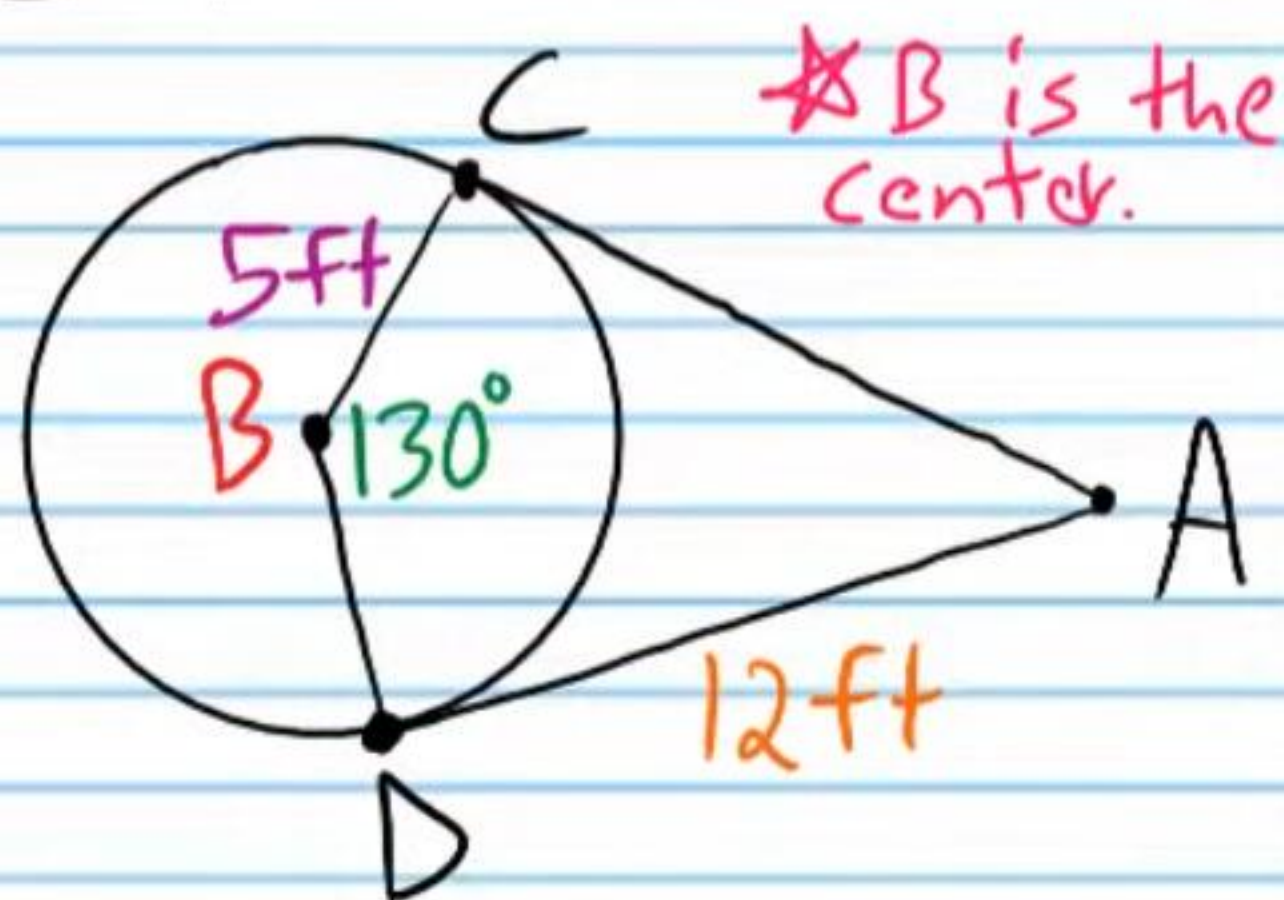


If C and D are points of tangency, find the following values.

(20) X



(21) i) AB & ii) $m\angle A$.

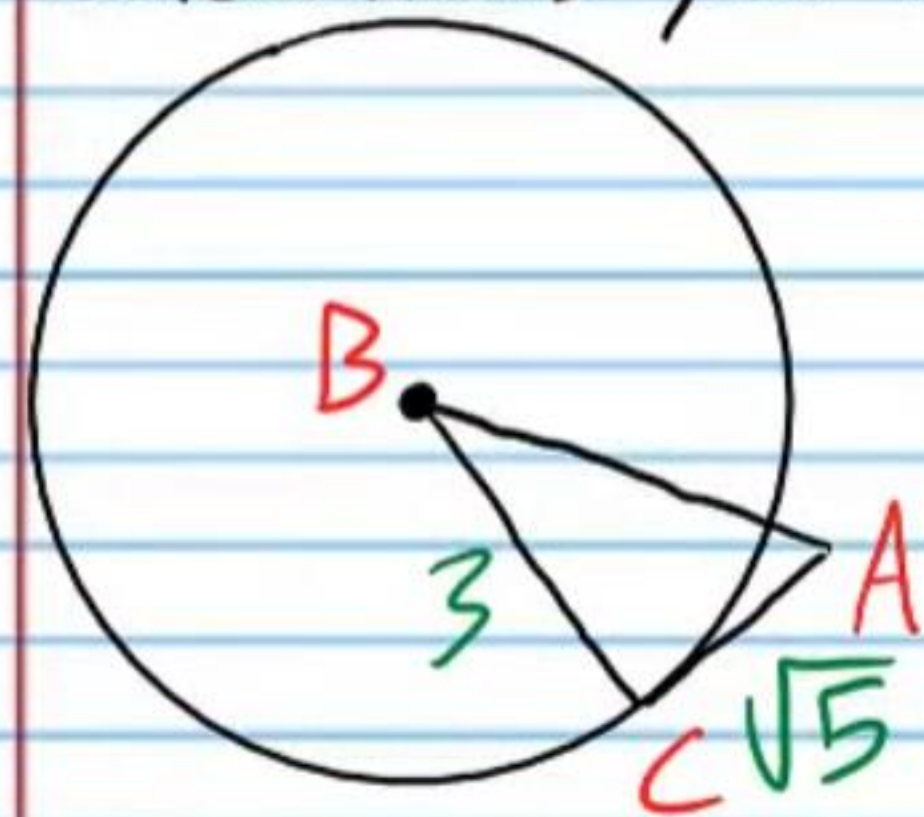


(22) If M is the center of the circle below, find $m\angle IMJ$. You must use the π button on your calculator. Round to the nearest hundredth.

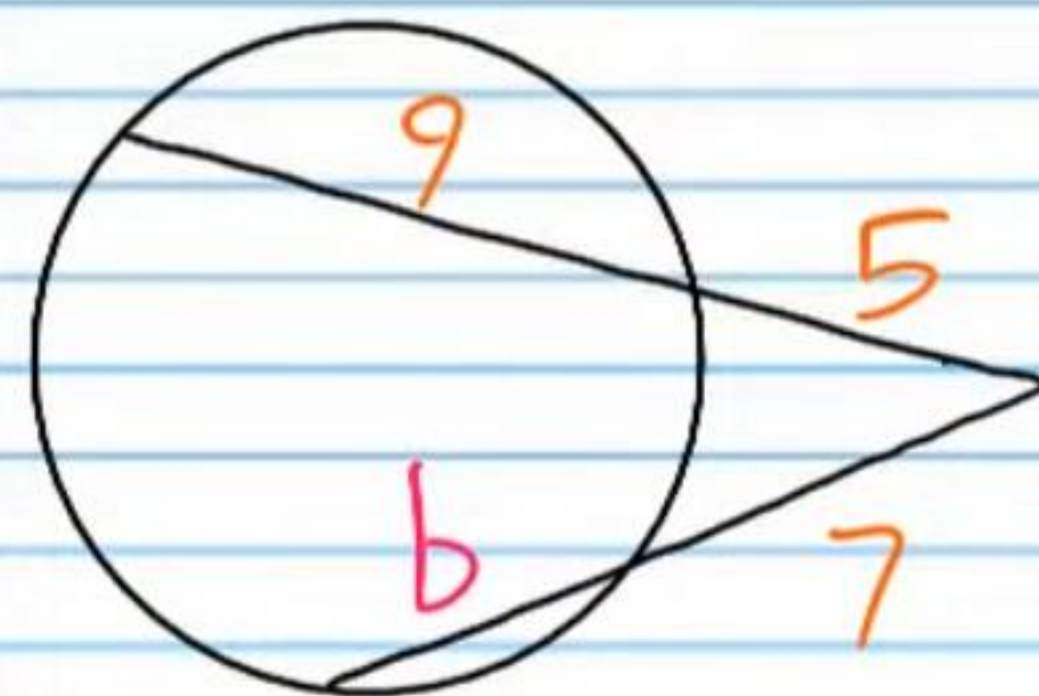


23 Find an equation of the circle containing a diameter with endpoints $(1, 2)$ and $(-3, 4)$.

24 If $\overline{BC}$ is a radius, and $AB = \sqrt{14}$, is the line containing $\overline{AC}$ tangent to $\odot B$? Show the calculations you use to determine tangency.



25 Find b .



Geometry Test V Answers

① $(x-1)^2 + (y-4)^2 = 4$

② $m\hat{M}P = 172^\circ$

③ $m\hat{A}B = 85^\circ$

④ No. $\overline{BC}$ bisects $\overline{AD}$ but is not perpendicular to it, so we cannot use the Chord Perpendicular Bisector Theorem.

⑤ $m\hat{A}B = 62^\circ$

⑥ $m\angle 3 = 55^\circ$

⑦ $m\angle D = 22^\circ$

⑧ We cannot use the Perpendicular Diameter Theorem because $\overline{QP}$ is not a diameter of the circle, so we do not have enough information to find the values.

⑨ $m\angle QM = 140^\circ$

⑩ $QS = 12$

⑪ $y = 2$

⑫ $x = 25/3$ or $8\frac{1}{3}$

⑬ $m\angle B = 61^\circ$

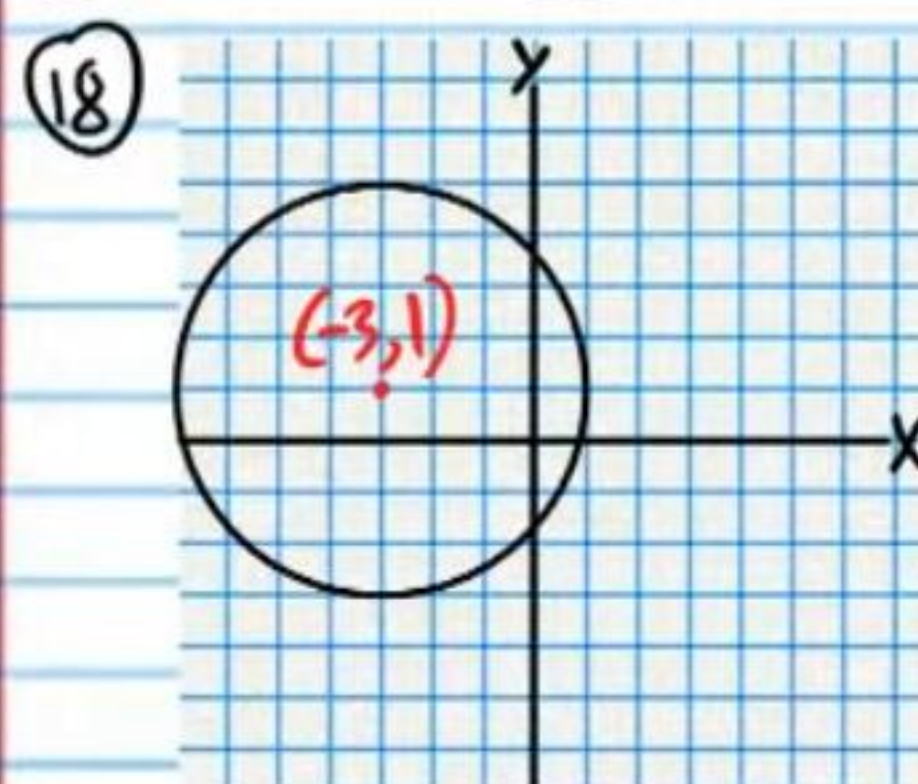
⑭ i) $m\hat{H}KI = 154^\circ$

ii) $m\angle HKJ = 180^\circ$

⑮ $\ell\hat{L}N = 21.8 \text{ cm}$

⑯ 16.9 in

⑰ $m\angle 1 = 38^\circ$



⑲ $AC = 5$

⑳ $x = 2$

㉑ i) $AB = 13 \text{ ft}$

ii) $m\angle A = 50^\circ$

㉒ $m\angle IMJ = 69.9^\circ$

㉓ $(x+1)^2 + (y-3)^2 = 5$

㉔ Yes. $3^2 + (\sqrt{5})^2 = (\sqrt{14})^2$
 $9 + 5 = 14$
 $14 = 14 \checkmark$

㉕ $b = 3$